

V současnosti pracuji na další konceptualizaci a testování modelu založeného na článku [The effects of alternative wage regimes in a monetary union: A multi-country agent-based stock-flow consistent model \(Caiani, Catullo, Gallegati\)](#). S tímto modelem, včetně přiloženého kódu, jsem se podrobně seznámil. Prostřednictvím svého vedoucího (Ph.D. Jiří Kukačka) jsem se spojil s hlavním autorem modelu Alessandrem Caianim a na základě jeho doporučení jsem částečně upravil svůj původní přístup. Autor uvedl, že můj přístup by měl v principu fungovat, zároveň však navrhl několik technických úprav i drobné změny v konceptuálním pojetí modelu. Těžiště mé práce nyní spočívá v experimentování s modelem a v jeho ladění. Domnívám se, že rámcové konceptuální ukotvení modelu mám již hotové, nicméně zbývá jej plně implementovat v kódu (Python) a získat z něj stabilní a ekonomicky smysluplné výsledky. V současnosti mám z velké části napsanou kapitolu literární rešerše, kterou pravděpodobně ještě doplním o několik citací a menší úpravy textu. Na vlastní psaní zbývajících částí práce (metodologie, výsledky, závěr) se zaměřím ve chvíli, kdy budu mít k dispozici funkční a spolehlivou verzi modelu.

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NÁZEV PRÁCE V ČEŠTINĚ

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Literature Review and Conceptual Background

This chapter presents a review of the relevant literature on Macroeconomic Agent-Based Modelling and the Credit Creation Theory of Banking. Furthermore, it examines studies, that explore bank behaviour in modern economies and its implications for economic growth and stability.

0.1 Macroeconomic agent-based models

Agent based computational economics (ACE) positions itself as an alternative to dynamic stochastic general equilibrium models (DSGE) (Dilaver, Jump, Levine, 2018). Different theoretical foundation of ACE models allows modeller greater freedom in specifying the underlying structure of the system being modelled and consequently allows for more realistic representation of the system in the model. As opposed to DSGE, ACE literature mostly disregards the notion of the so-called representative agents, which are “typical decision-makers of a certain type” and which influence each other only indirectly, through the aggregate variables of the model, in favour of (finitely) “many” heterogeneous agents which can interact also directly with each other (Axtell Farmer, 2022). In ACE models, such agents are typically boundedly rational, making decisions based on heuristics of varying complexity, and operate in environments characterised by information asymmetries. Models employing such dynamics generally cannot be expected to converge to equilibrium in the sense that macroeconomic models employing “more conventional” frameworks do (Dosi Roven-
tini, 2025). Of course, some sort of Statistical equilibrium(s)* is likely to emerge in the dynamics of ACE models, but anything like a Walrasian equilibrium should not be expected (Delli Gatti et al. (2018)). Hence, in the ACE literature, the focus is usually on studying the system’s dynamics and matching the qualitative properties of the system being modelled, i.e. replicating stylized facts. At the same time, several recently developed macroeconomic ACE models achieve out-of-sample

predictive performance comparable to state-of-the-art DSGE models (Farmer 2025), and the field is shifting from “roughly matching empirical regularities” toward formal estimation of model parameters (Dyer et al., 2024). Since the methodological framework of ACE is suitable for representing, potentially in detail and with relative ease, systems in which agent heterogeneity, bounded rationality and their mutual interactions play a crucial role, it can be used to formally explore questions that would be hard to even ask using standard macroeconomic modelling techniques (Farmer, 2025). It is argued that these features make ACE models well-suited for modelling modern monetary economies, where agent heterogeneity, bounded rationality and mutual interactions at the micro level play a significant role in shaping the system dynamics (Caiani et al., 2016) and can generate outcomes that would seem “nearly impossible” if we do not account for this intrinsic complexity (Farmer Foley 2009; Stiglitz 2018). This fact is indeed well realized in the growing field of ACE and is increasingly being recognized within the economics profession (Rosser 2021; Haldane Turrell 2017; Borsos et al. 2025). We conclude this section by noting that, since macroeconomic agent-based models are best calibrated on micro-level data— which, unlike aggregate time-series data, are abundant for most modern economies (Farmer, 2025)—and can directly benefit from increases in computing power (Axtell Farmer, 2022), they constitute a particularly promising line of inquiry for macroeconomics.

0.2 On Money creation in the modern economy

Over the past century, the economics profession’s consensus on money¹ and banks—especially on the role of banks in money creation in modern economies—has shifted several times. Because this evolution has been somewhat circular, and to this day, the explanations provided in leading economics textbooks generally differ (Carlin Soskice, 2024; Mankiw, 2021) - we elaborate on this development here. Strictly speaking, the core insight - already recognised in the early twentieth century (Werner, 2014) - that new means of payment, in the form of bank deposits (bank IOUs), are created when a bank extends a loan, is correct (Werner 2014a, 2014b; McLeay, Radia Thomas, 2014a,b; Deutsche Bundesbank, 2017). Since in the contemporary monetary economies, bank-deposit money created through lending constitutes the dominant share of money supply (broad money), with the remainder consisting of money created by central banks, this insight is significant (McLeay, Radia Thomas, 2014).

Most recently, it was brought to the attention of the broad economics profession

¹Understood here in the broad sense as the dominant means of payment in the economy.

again, by the 2008 financial crisis (previously this view of banking sector was popular particularly with the post-Keynesian*** and ‘Austrian’ economists (Werner, 2014)) before which the dominant approach, used in modelling the banking sector was the intermediation of loanable funds (ILF) (Jakab Kumhof, 2015). This line of thinking was popularized in the second half of twentieth century by an influential (but misleading) interpretation of the role of banks in Tobin, 1963 (based on the previous work of Gurley and Shaw (1955, 1956)) as argued by Jakab Kumhof, 2015.

Within the ILF view, banks must first accumulate savings in order to be able to lend. In reality, although each newly created loan is immediately offset in the bank’s accounting system by a corresponding liability in the form of a deposit; this is just a formal operation ensuring accounting consistency, and no pre-existent “real” savings need exist for a bank to perform this procedure (Werner, 2005). As Adair Turner (2018), former head of the UK Financial Services Authority, puts it, “they [banks] create credit and money *ex nihilo*”. It therefore follows that banks can create new loans—and thus new bank money, which in modern economies constitutes the main component of broad money—against the debtor’s promise to repay later (usually secured on the debtor’s assets). In the “short run” banks are constrained in this only by their profitability and solvency considerations, rather than external constraints such as availability of “loanable funds”, or the central bank reserves (Jakab Kumhof, 2015).

Here we abstain from a more detailed discussion of the credit-creation “theory” of banking, which is covered in detail, for example, in McLeay, Radia and Thomas (2014a, 2014b), and instead focus on its implications for the dynamics of the “system in which it is being implemented”. As noted above ILF theory assumes that bank loans are only made possible by the prior accumulation of the bank deposits, i.e. that the funds used by banks to lend must be first somewhat created in the real economy. This stands in stark contrast to the credit-creation view, which asserts that the loan can be created just against the obligation of the debtor. The latter view permits much less smooth dynamics of the economic system, as it allows for potentially large “jumps” in the supply of credit. This aspect is discussed at length and tested in the framework of two otherwise identical DSGE models in Jakab Kumhof, (2015). Indeed, the insights derived from models implementing ILF and from models implementing financing through money creation (hereafter FMC, following terminology of Jakab and Kumhof, 2015) differ substantially.

To conclude, we note that, in contrast to the DSGE literature (with some notable exceptions such as Beneš Kumhof, 2012, and Beneš, Kumhof Laxton, 2014), the majority of macroeconomic ACE models implement the FMC view of bank-

ing. We hypothesize that this is due to the fact that they generally build on the post-Keynesian framework.²

0.3 On Banks and economy

It is a well-known fact that banks' willingness (or lack thereof) to extend credit exerts a significant influence on the dynamics of the business cycle (Fisher, 1933; Minsky, 1986). It is also well established, that credit markets are characterised by rationing, in the sense that the market outcome is determined by the "short side" rather than by the intersection of supply and demand, and that the credit is typically rationed (Jaffe Russel (1976); Stiglitz Weiss, (1981))³. Since banks are profit-driven institutions, it follows that the most profitable allocation (from the bank's perspective), not necessarily the most socially optimal one, is to be expected (Stiglitz Weiss, 1981; Werner, 2012).

The market imperfection stemming from this set-up that we are particularly interested in is that small firms are typically credit rationed (Mc Namara et al., 2020; Jin Zhang, 2019), and this appears to be more pronounced in some contexts than in others (Mkhaiber Werner 2021). In particular, banking sectors that are highly concentrated and dominated by large banks tend to discriminate against small firms in credit provision, whereas banking sectors composed of many small, locally operating banks do not appear to exert such negative effects on the credit conditions of small firms (Mkhaiber Werner, 2021). A prominent example of the latter configuration is Germany: its dense network of local savings and cooperative banks has long been the primary provider of credit to small and medium-sized enterprises (Behr Schmidt, 2015; IMF, 2011). This feature of the German banking sector, alongside other institutional factors, is frequently cited as one of the contributors to the country's long-standing economic success (Behr Schmidt, 2015; IMF, 2011; Mear Werner, 2020).

We tentatively conclude from this example that a combination of small firms and "small finance" may be desirable with respect to both the growth potential and the stability of the economy. To explore this hypothesis, we attempt at embedding this framework in a fully fledged stock-flow consistent macroeconomic agent based model, which we describe in the next section.

²Many prominent macroeconomic ACE models are post-Keynesian in spirit, frequently combining agent-based techniques with stock-flow consistent frameworks etc. (Dosi et al., 2010; Dosi et al., 2013; Caiani et al., 2016; Dosi Roventini, 2019.)

³They show this in a context of ILF.

Methodology

To attempt at providing tentative answer to the above raised question, we employ the stock-flow consistent macroeconomic agent based model developed in Caiani, Catulo Ricetti (2019).

The model is available in open-source form the authors web page, which makes it easily extensible.⁴ Further, the model is calibrated to the conditions of the generic European economy, which we consider useful since we want to work in a "broadly European context" regarding our analysis.

The stock flow consistency in the context of this model means, that all of the financial flows are being exactly tracked in the model economy. Since each economic unit is equipped with its own balance sheet, erroneous handling of the funds in the model economy is excluded by design.⁵ Further reasons for adoption of this framework include the advantages of ACE models described in the previous chapter.

0.4 Model description and modification

We have decided at keeping the model in a two country setup. We chose this baseline configuration in order not to distort the original model calibration. (...)

⁴The model is implemented in Python, a programming language in which the author of the thesis is proficient, which further lowers the cost of adapting and extending the code.

⁵For a comprehensive treatment of stock-flow consistent modeling, see the seminal work on the topic by Godley and Lavoie (*Monetary Economics: An Integrated Approach to Credit, Money, Income, Production, and Wealth* (2012)).

Bibliography

Author, A. (2020). *Title of the book or article*. Publisher or Journal.

Appendix Title

You can put additional tables, figures or derivations here.